IMY 220

Practical 3

React

Due: Friday, 14 August 2026 at 08:20

The submission instructions are available on ClickUP. Any deviation from these instructions will result in a 10% deduction from your mark.

Instructions

You will create a simple React application that displays a PhotoShare feed containing several social media posts.

You may not use loops or array iteration methods to generate the posts. This includes `for`, `forEach()`, `do...while`, `while`, `map()`, `filter()`, and `every()`.

You have been provided with boilerplate code that you can use to complete this practical.

Complete the practical using functional components, not class components.

The supplied CSS should be used. You may customise the colours if you wish, but the overall layout should remain similar.

Ensure that your student number is included as a comment in the `index.html` file. Replace `uXXXXXXX` with your student number.

Section 1

All your code will be written inside the `<script>` tag of the `index.html` file. You do not need to set up a React project.

Create a component called `App`. This component will serve as the starting point of your practical.

For now, the component should contain only an <h1> element displaying the following heading:

PhotoShare Feed

Once the component has been created, render the `App` component inside the following element:

Section 2

Create another component called `PostCard`.

The component should receive the following props:

- `username`
- `caption`
- `comment`

Display the username inside an <h2> element.

Display the caption inside a <p> element.

The component should receive the `comment` prop, but the comment should not be displayed initially.

Section 3

Inside the `PostCard` component, implement functionality that allows the user to show and hide the comment.

When the comment is hidden, the button should display:

Show Comment

When the comment is visible, the button should display:

Hide Comment

Clicking the button should toggle the visibility of the comment.

Each `PostCard` component should manage its own comment visibility state independently.

Section 4

Reuse the `PostCard` component to display the following three posts:

Username	Caption	Comment
`@sarah`	Studying React today!	React is actually quite fun!
`@john`	Finished Practical 3.	Great job!
`@leanne`	Sunset over campus.	Beautiful photo!

Create the three `PostCard` components manually. Do not use a loop or an array iteration method to generate them.

At the end of this section, the page should display three reusable `PostCard` components.

Section 5

Allow the user to change the heading of the page.

Add a text input above the heading.

The text input should initially display:

PhotoShare Feed

The text input should be controlled using React state.

As the user types into the text input, the page heading should immediately update to display the new value.

Example Layout

[PhotoShare Feed]

PhotoShare Feed

@sarah

Studying React today!

[Show Comment]

@john

Finished Practical 3.

[Show Comment]

@leanne

Sunset over campus.

[Show Comment]

After clicking **Show Comment**:

@sarah

Studying React today!

React is actually quite fun!

[Hide Comment]

Submission Files

- `index.html`

Submission Instructions

ZIP all submission files using a tool such as 7-Zip or WinRAR.

Name the ZIP file:

p3-{student_number}.zip

Replace `{student_number}` with your student number.

Submit your ZIP file before 08:20 to avoid a late submission.